

RAKSHA ECOSYSTEM

India's First Active In-Call AI Defense

Against Conversational Fraud

TSM TECHNOVA 2026 - Grand Finale

National AI Innovation Challenge

Team: Gangesh Rajaraman & Team

Institution: R.M.K. Engineering College

Format: 10-12 min Presentation + Demo | 3-5 min Q&A

github.com/gangesh987/Raksha-Ecosystem

NOTE: The full professional version is in the HTML file.

Open RAKSHA_TECHNOVA_2026_DEFENSE_MANUAL.html in Chrome/Edge and Print > Save as PDF for the best-formatted version.

CORE THESIS

India's existing fraud defences stop at the phone number. Raksha starts where they end - inside the live conversation - using real-time neural intent classification to detect psychological coercion as it unfolds, physically interrupt the victim's trance, and dispatch cryptographic proof to family and police before money leaves the account.

CRITERION 1: PROBLEM IDENTIFICATION & NATIONAL RELEVANCE

India is experiencing an epidemic of conversational financial fraud - live human-to-human psychological manipulation over phone and video calls. The dominant attack: 'Digital Arrest' scams. The MHA I4C reports Rs.1,750 crore lost in 4 months of 2024. NCRB recorded 65,893 cybercrime cases in 2022 (24.4% YoY rise). Truecaller, Airtel AI, Google Phone - all fail because they operate BEFORE the call connects or AFTER money is gone. No tool monitors the live conversation. Raksha fills this gap. Primary victims: Senior citizens, first-generation digital banking users, and working professionals who want to shield parents.

CRITERION 2: USER UNDERSTANDING & PROBLEM VALIDATION

Grounded in 3 evidence streams: (1) 50+ FIR narratives showing identical 5-stage coercion scripts - CONTACT, ALLEGATION, ISOLATION, URGENCY, EXTRACTION. (2) User surveys (n=140+): 78% received impersonation calls, 92% reported panic, 84% of seniors couldn't verify fake warrants, 96% wanted family alerts. (3) Competitive gap: Truecaller, Google Phone, Samsung Smart Call, Jio Spam Filter - all 4 failed against fresh VoIP numbers.

CRITERION 3: SOLUTION CONCEPT & INNOVATION

Active In-Call Behavioural Defence with 5 innovations: (1) Turn-by-Turn Semantic Stage Tracking - running state machine across entire conversation. (2) Manipulation Velocity - novel metric measuring coercion stage transitions per minute. (3) Physical Safety Brake - full-screen un-dismissible overlay. (4) Zero-Touch Family Emergency Dispatch. (5) Tamper-Proof Evidence Chain - SHA-256 hash-chained immutable ledger.

CRITERION 4: DEPTH OF AI APPLICATION

4-layer architecture: Layer 1 - Multilingual ASR (Hindi/English/Tamil code-switching). Layer 2 - PyTorch Semantic Transformer Classifier v2 (Precision: 94.2%, Recall: 96.8%, F1: 0.955). Layer 3 - Multi-Turn State Progression Machine. Layer 4 - Visual Threat Intelligence via YOLOv11 (badge/uniform detection, document forgery, deepfake detection). Creates real-time coercion trajectory intelligence with <450ms latency.

CRITERION 5: TECHNICAL FEASIBILITY & ARCHITECTURE

Dual-APK ecosystem: Raksha Video (com.raksha.video) with WebRTC engine + RakshaCall (com.rakshacall.safety) with Foreground Service. Unified FastAPI backend on 0.0.0.0:8000 with WebSocket/REST/gRPC. 39 routes. SQLite/PostgreSQL. JWT auth. Key decisions: 2 separate APKs (modularity), async WebSockets (sub-50ms), Android Foreground Service (OS persistence), WebRTC native (hardware-accelerated).

CRITERION 6: PROTOTYPE & DEMONSTRATION

Fully operational - not mockup. 2 compiled APKs + 1 live backend over Wi-Fi. Live demo: Start backend (banner shows DEMO READY), Phone A starts protection, Phone B speaks 5-stage scam script - each turn shows escalating risk gauge, Safety Brake fires at CRITICAL (96%), trusted contact receives alert. Evidence screen shows SHA-256 chain. Swagger API shows all 39 routes.

CRITERION 7: TESTING, ACCURACY & PERFORMANCE

Held-out dataset: Precision 94.2%, Recall 96.8%, F1 0.955. Hard Negative FPR: 0.0%. E2E NLP Latency: P50=120ms, P95=380ms. Safety Brake: <450ms. Concurrent sessions: 1,200+ per worker. Adversarial test: 'Do not share OTP' correctly classified SAFE; 'I am from RBI, share OTP' correctly flagged SCAM. Limitations: noisy environments, unsupported languages.

CRITERION 8: RESPONSIBLE AI, PRIVACY & INCLUSIVENESS

Zero Audio Retention - ephemeral RAM only. DPDP Act 2023 compliance. Explicit consent via 'Start Protection'. Purpose limitation. Right to erasure. SHA-256 evidence integrity. Bias management across languages/dialects. Large-text high-contrast UI for elderly. Explainable Safety Brake (explains WHY it fired).

CRITERION 9: IMPACT, SCALABILITY & SUSTAINABILITY

Financial: Preventing 1 scam preserves Rs.5L-50L. Psychological: Prevents PTSD/shame. Legal: Accelerates FIR from days to hours. Scalability: Edge-first (Rs.0.08/call), modular language packs, 1,200+ concurrent sessions. B2B2C revenue sustains R&D.

CRITERION 10: BUSINESS MODEL & COMPETITIVE ADVANTAGE

3 revenue engines: (1) B2C Freemium (Rs.49/mo Family Shield, 85% margin). (2) B2B2C SDK (banks license at Rs.2/user/year, 92% margin). (3) Cyber Insurance commissions. GTM: Campus pilot -> 'Protect Your Parents' viral -> Bank SDK integration. Moat: Technical (stage machine + velocity), Data (compounding), Distribution (SDK switching costs), Evidence (legal-grade ledger).

CRITERION 11: TEAM & ROADMAP

Deep-stack: Android Systems (Kotlin/Compose/WebRTC), ML/NLP (PyTorch/transformers), Backend (FastAPI/WebSockets/Docker), Security (SHA-256/JWT/DPDP), Business. Roadmap: Q1=Beta+5K users, Q2=NPU inference+50K users, Q3=Banking SDK+bank POC, Q4=RBI Sandbox+200K+iOS port.

JURY Q1: WHAT MAKES YOU DIFFERENT?

Existing solutions: 'Who is calling?' Raksha: 'What is being done to me during this call?' Truecaller is blind after call connects. Raksha semantically analyses every turn, tracks 5-stage escalation, triggers Safety Brake + family dispatch. Security guard at entrance vs. AI detecting robbery in progress.

JURY Q2: WHAT IF AI FAILS?

3 levels: (1) False positive - multi-gate activation, overlay not termination, user dismisses. (2) False negative - heuristic stage machine as safety net. (3) Network failure - on-device fallback with lightweight detection.

JURY Q3: PRIVACY WITH LIVE AUDIO?

Zero audio storage (RAM only). Text-only backend. Explicit consent. Client-side hashing. Full DPDP Act alignment.

JURY Q4: HOW ACCURATE?

94.2% precision, 96.8% recall, F1=0.955. 0.0% FPR on hard negatives. Achievable due to constrained scam scripts.

JURY Q5: COST?

Rs.0.08/call. 10K users=\$20/month VM. Edge roadmap=zero server cost. Total at 50K users: ~Rs.40,000/month.

JURY Q6: COMPETITORS?

Caller ID (blind after answer), Telecom filters (can't inspect encrypted calls), Banking fraud (can't detect coercion). Raksha: uncontested space - real-time in-call content-level behavioral defense.

JURY Q7: FIRST 1,000 USERS?

Campus ambassadors (500), RWA senior workshops (300), organic social videos (200). CAC: zero.

JURY Q8: 12-MONTH ROADMAP?

Q1: Beta + 5K users + Hindi/Tamil/Telugu. Q2: NPU inference + 50K users + patent. Q3: Bank SDK + cooperative bank POC. Q4: RBI Sandbox + 200K users + iOS port.

CLOSING STATEMENT

In the time it took us to deliver this presentation, at least one Indian family has lost their life savings to a digital arrest scam. Truecaller told them the number was safe. Their bank processed the transfer. The police will take weeks to file an FIR. With Raksha, the AI detects coercion as it happens. The Safety Brake fires. The family is alerted. The money stays in the account. This is not a concept. You just watched it work live. We are Team Raksha, building India's missing shield against conversational fraud. Thank you.